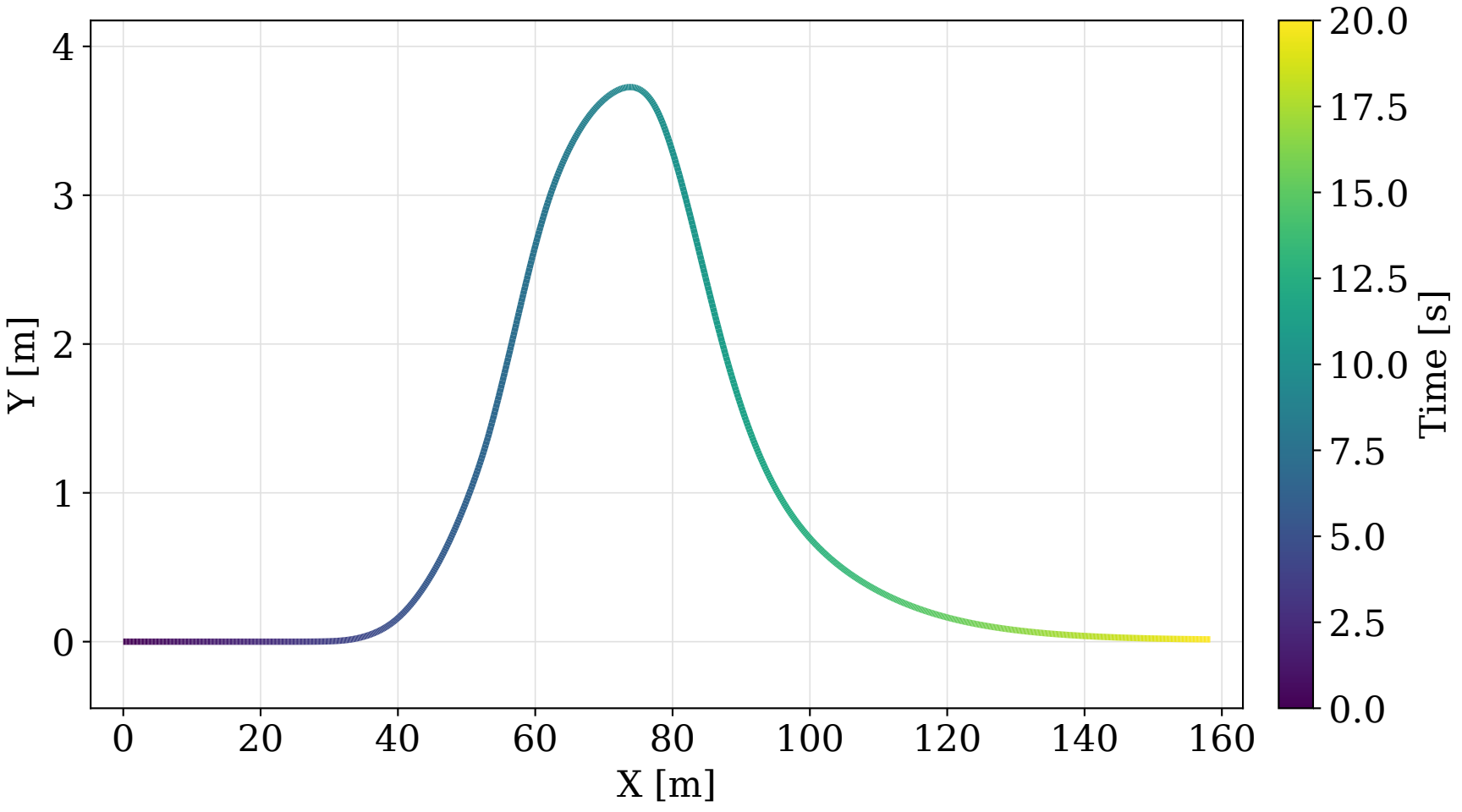
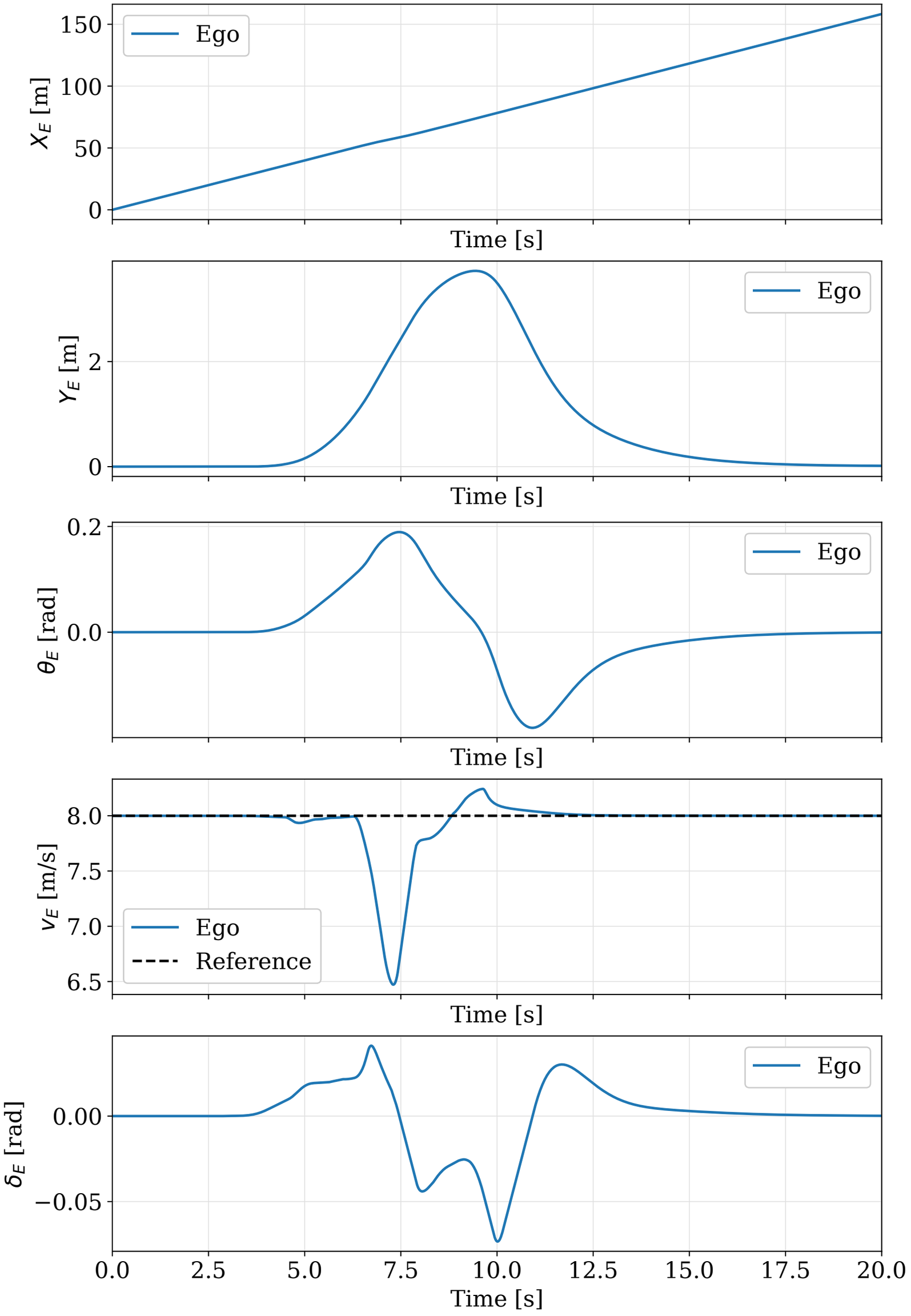
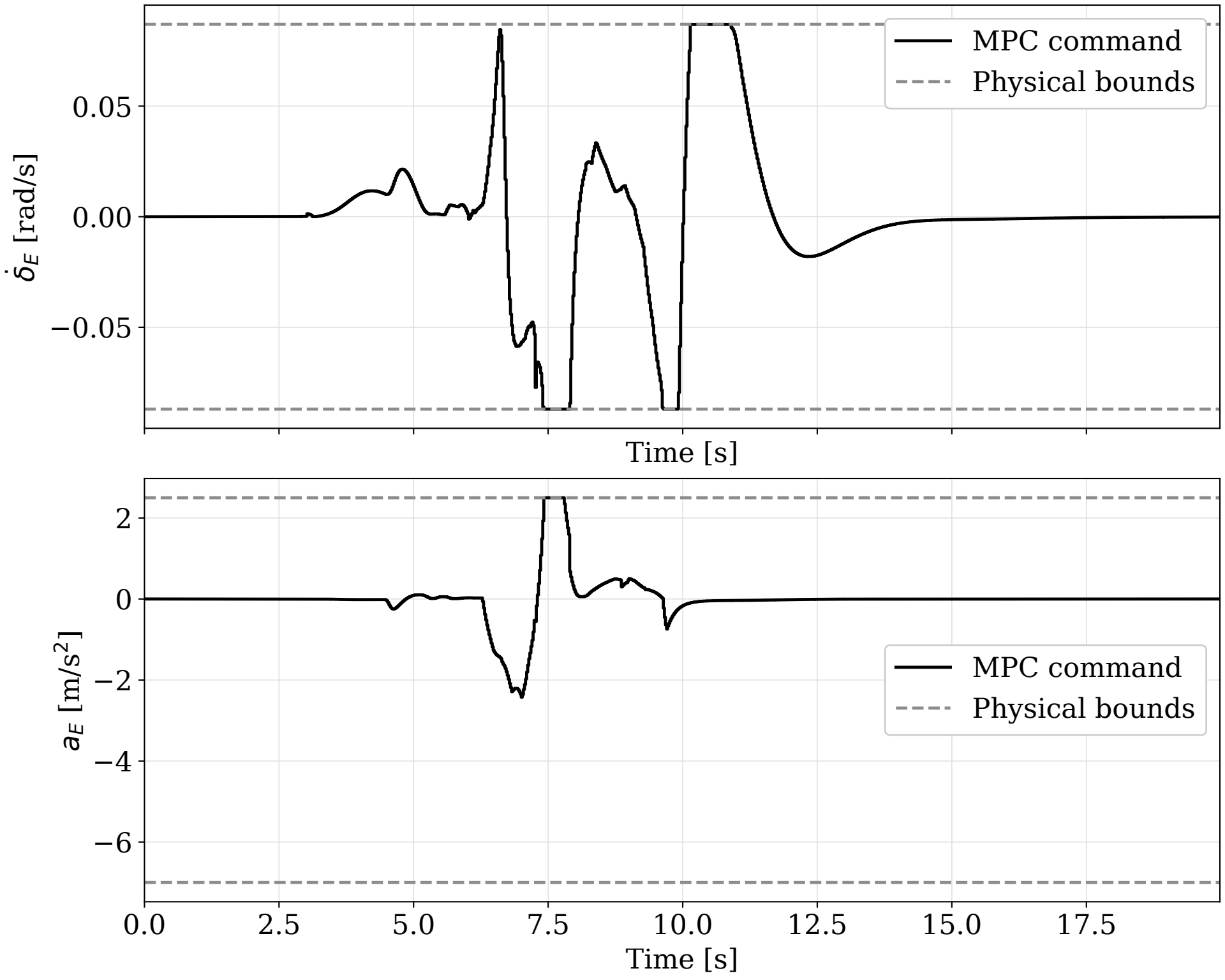
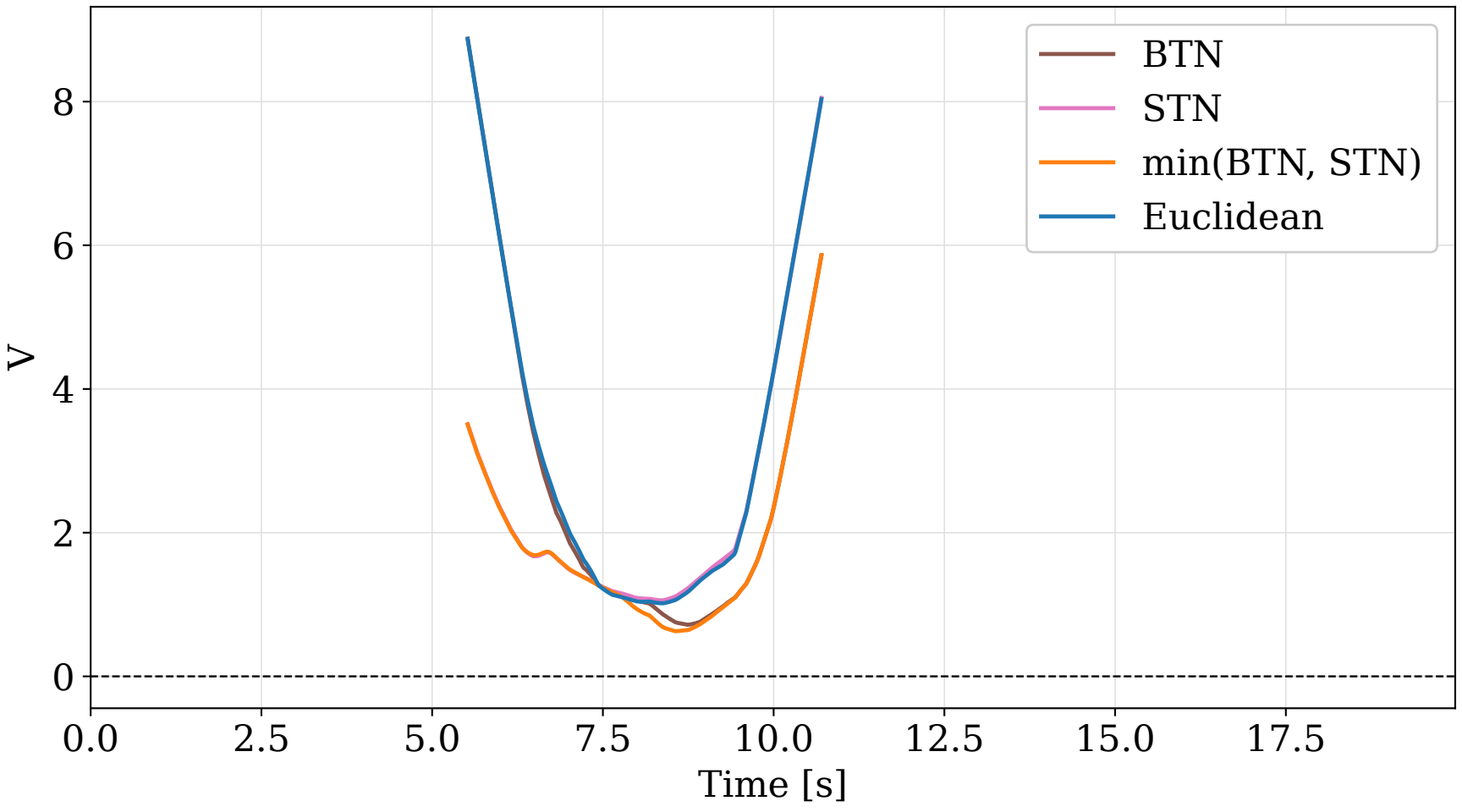
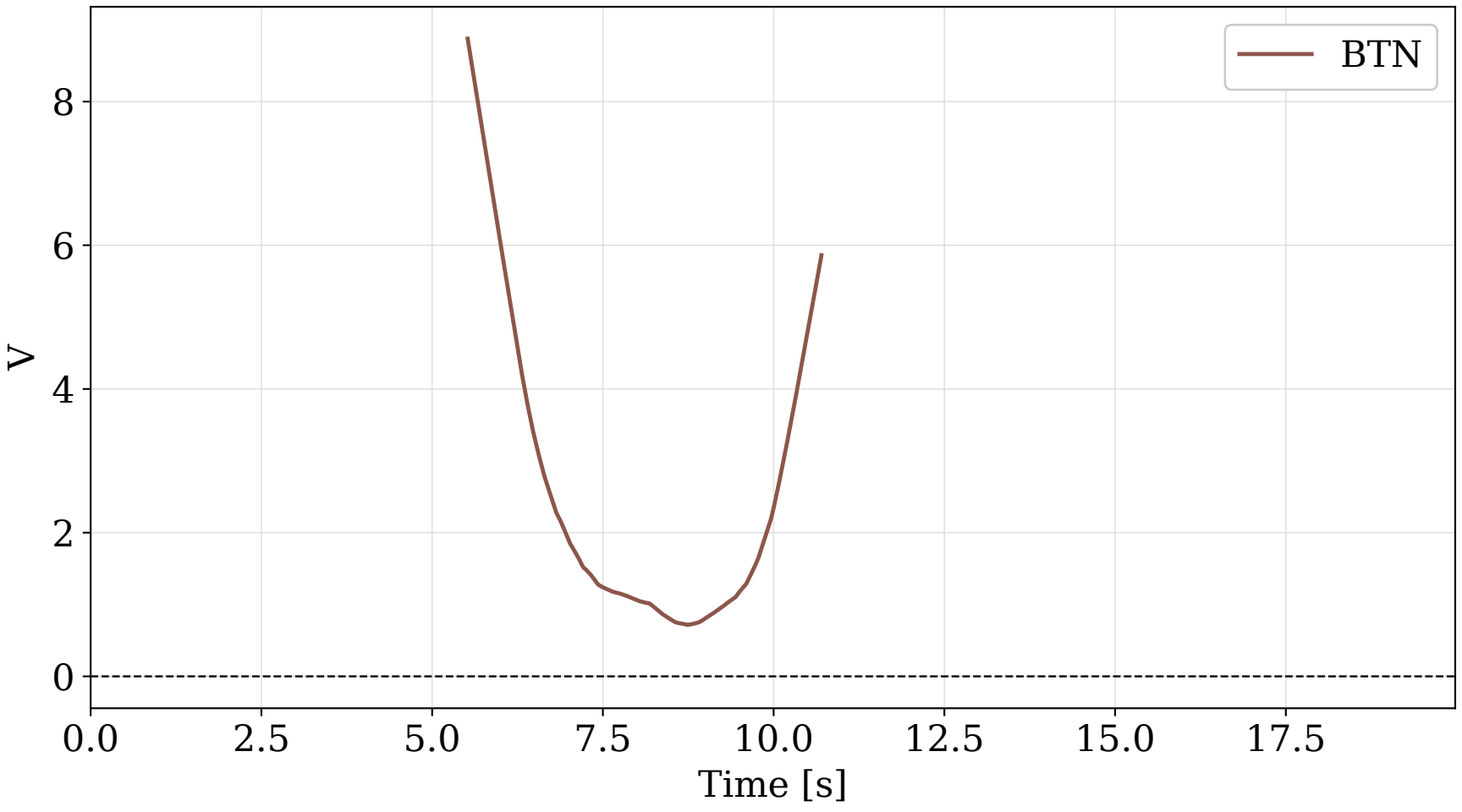
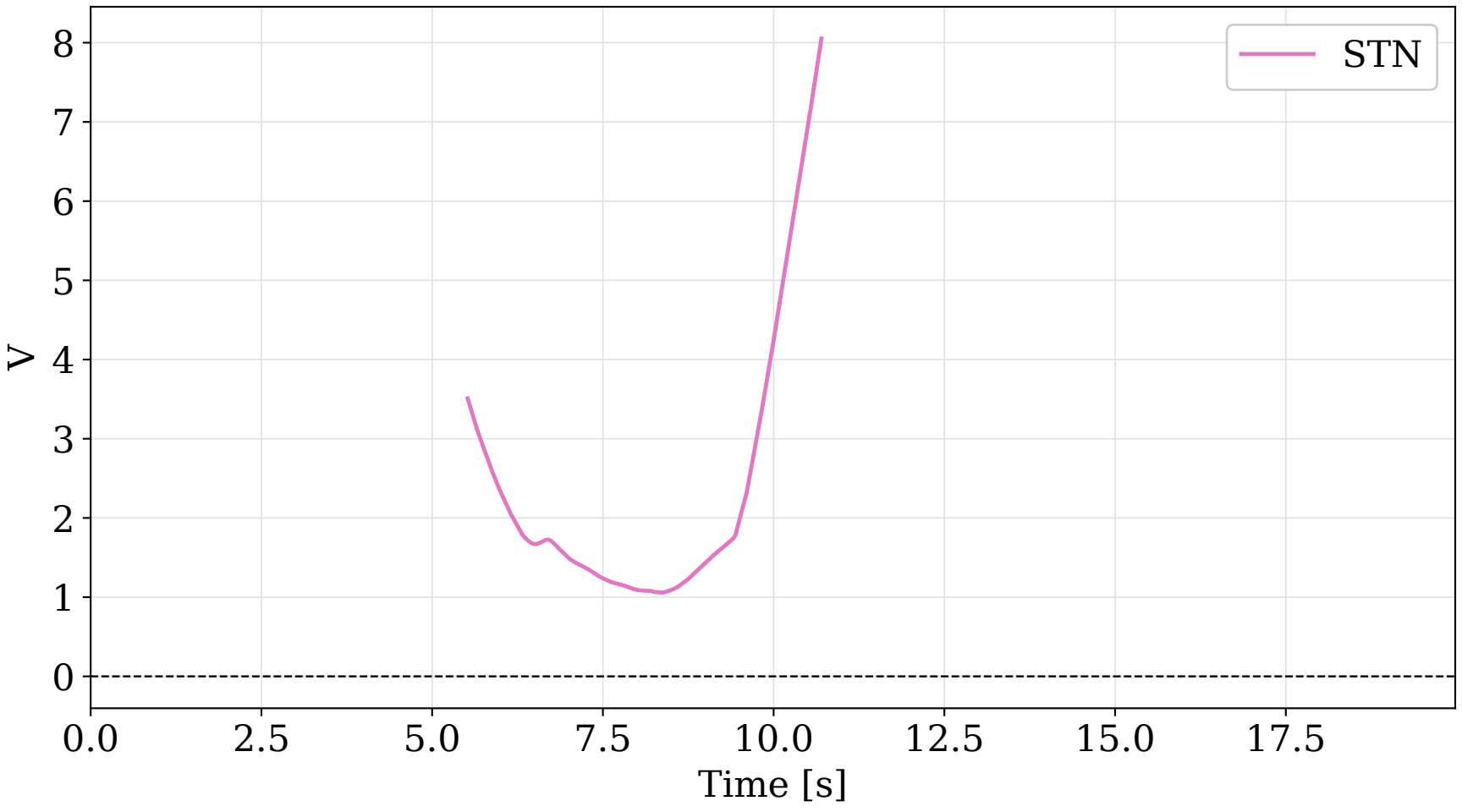

Y [m]
X [m]
Time [s]
0
1
2
3
4
0
20
40
60
80
100
120
140
160
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

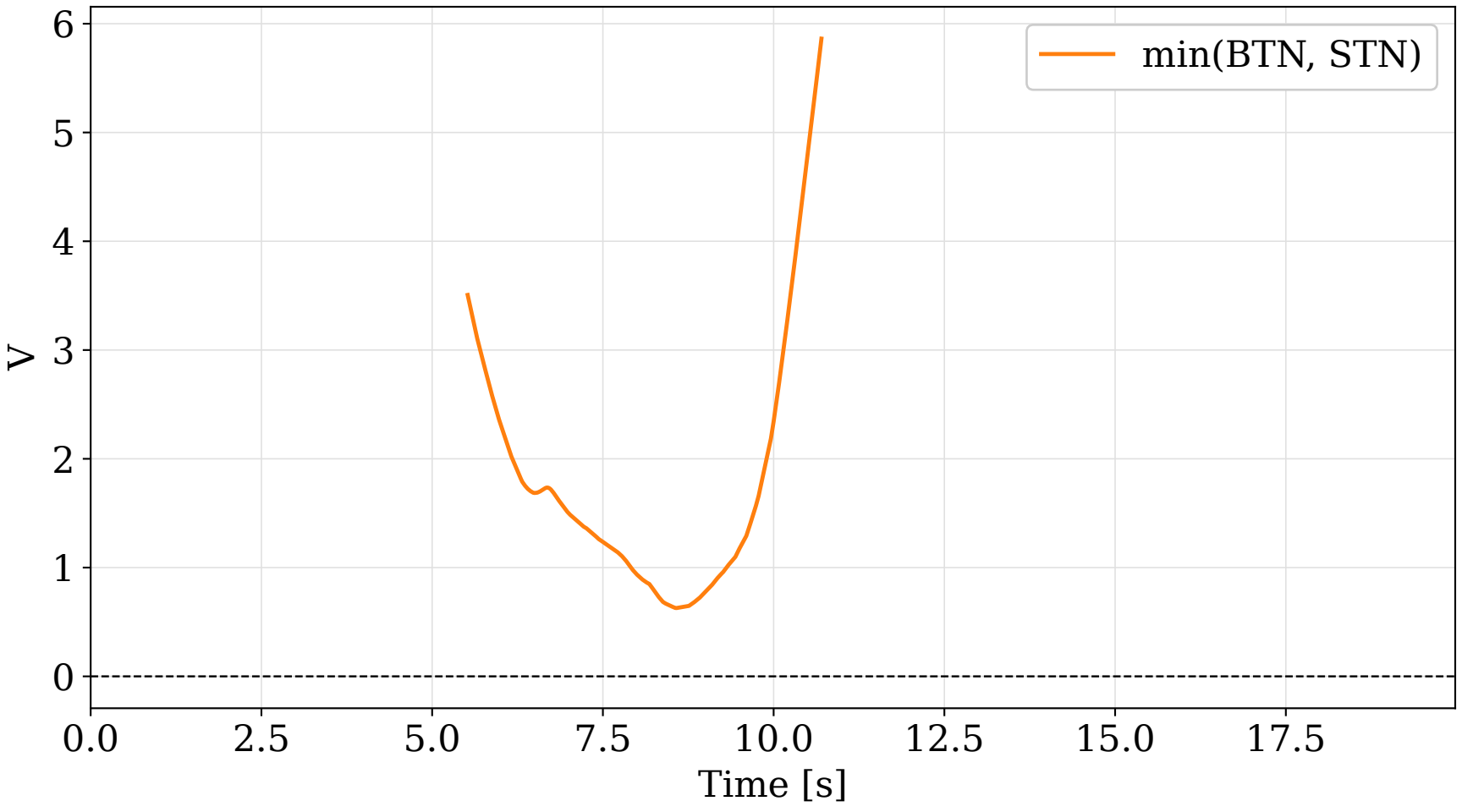

Ego
X_E [m]
150
100
50
0
Time [s]
Ego
Y_E [m]
2
0
Time [s]
Ego
θ_E [rad]
0.2
0.0
Time [s]
Ego
Reference
v_E [m/s]
8.0
7.5
7.0
6.5
Time [s]
Ego
δ_E [rad]
0.00
−0.05
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

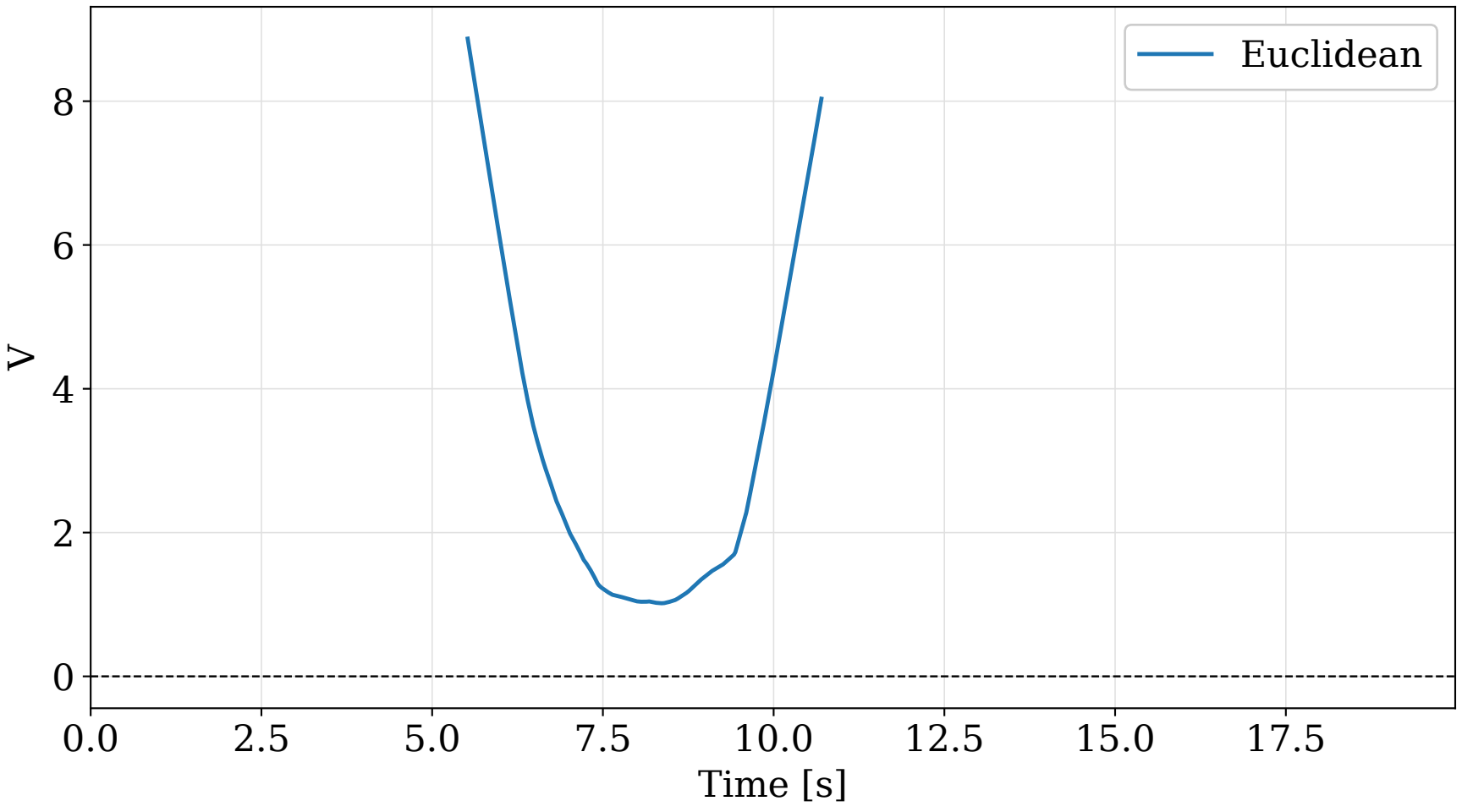

MPC command
Physical bounds
$\dot{\delta}_E$ [rad/s]
0.05
0.00
−0.05
Time [s]
MPC command
Physical bounds
a_E [m/s^2]
2
0
−2
−4
−6
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time [s]

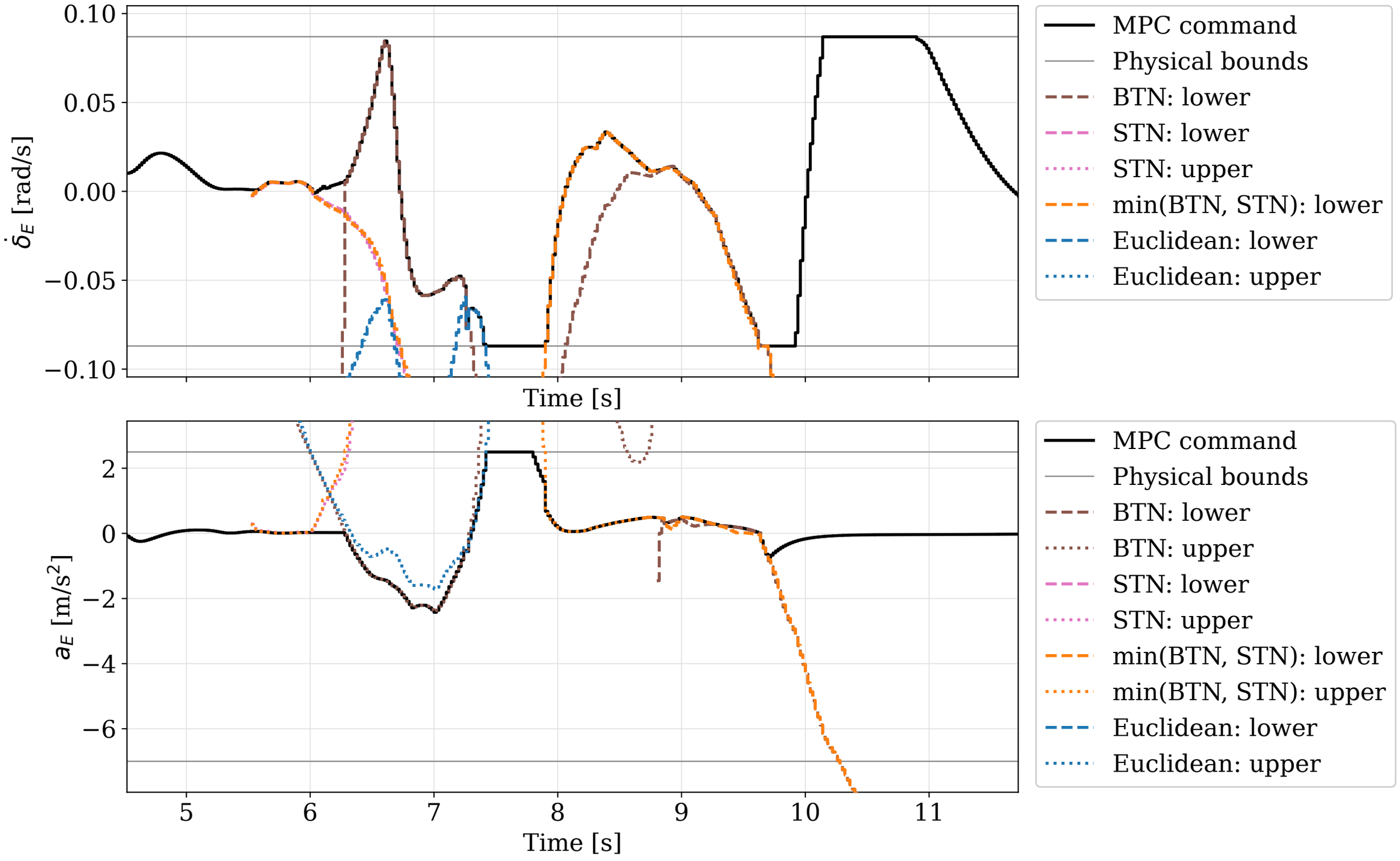

BTN
STN
min(BTN, STN)
Euclidean
V
8
6
4
2
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time [s]

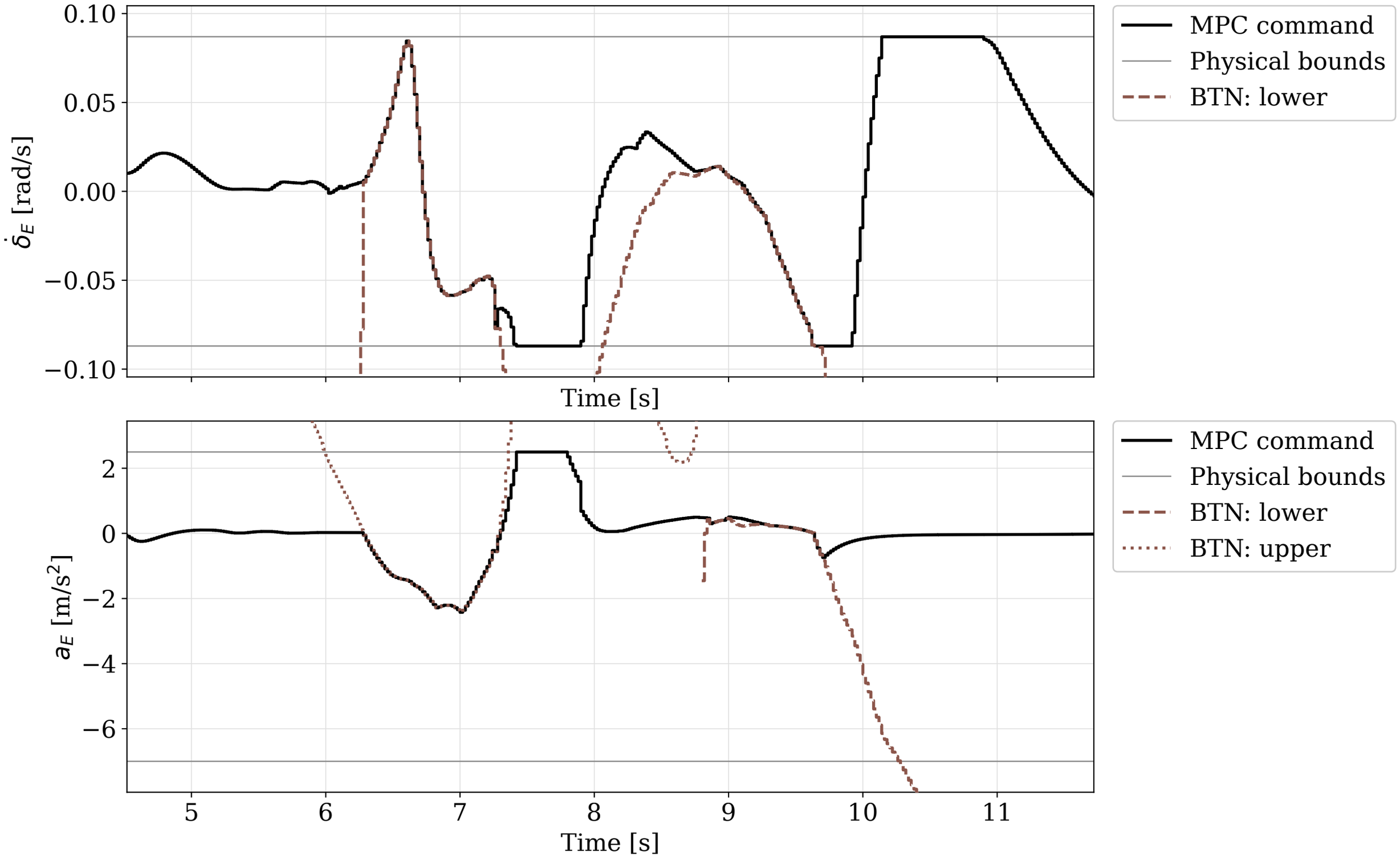

BTN
8
6
4
2
0
V
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time [s]

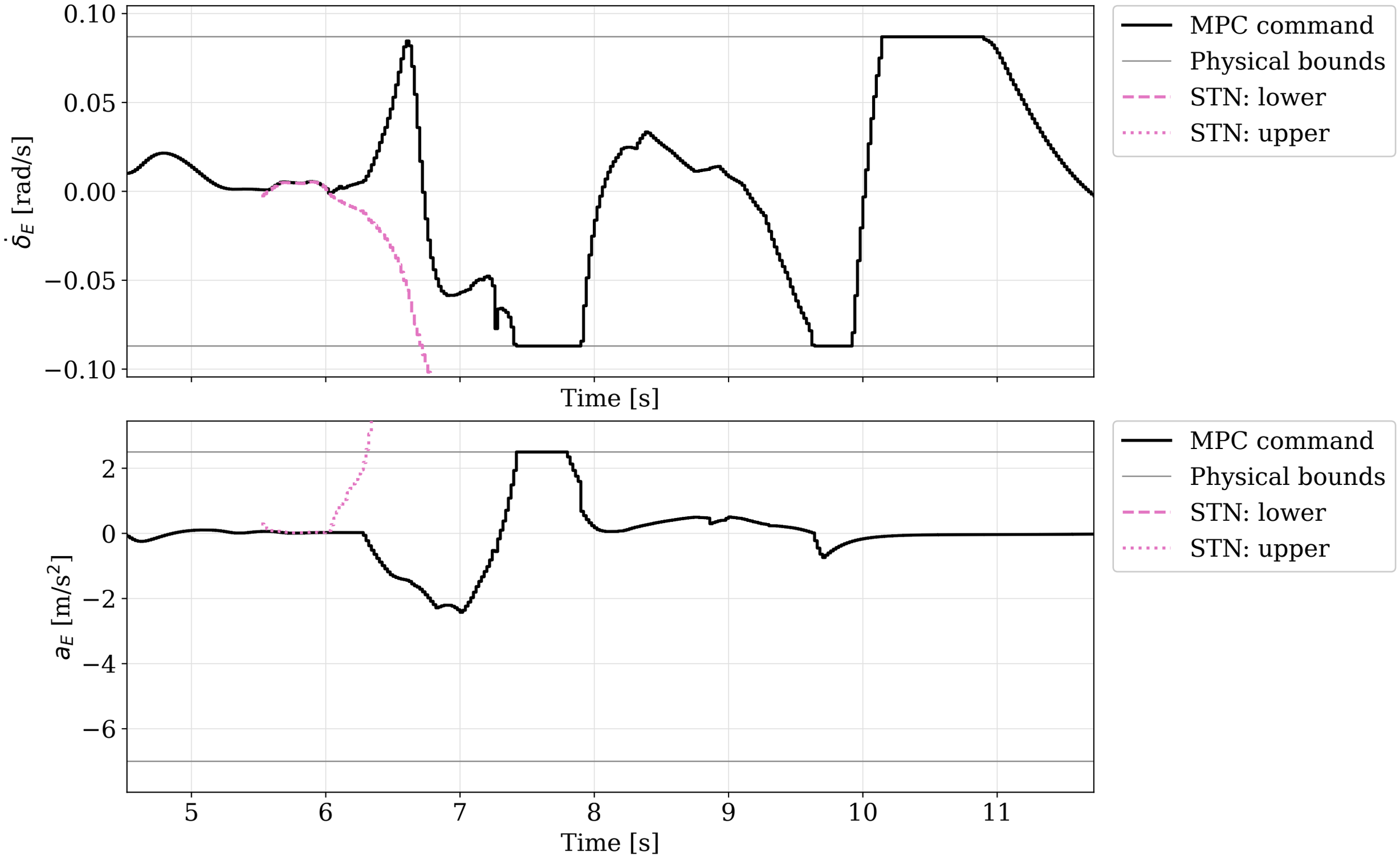

STN
V
8
7
6
5
4
3
2
1
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time [s]

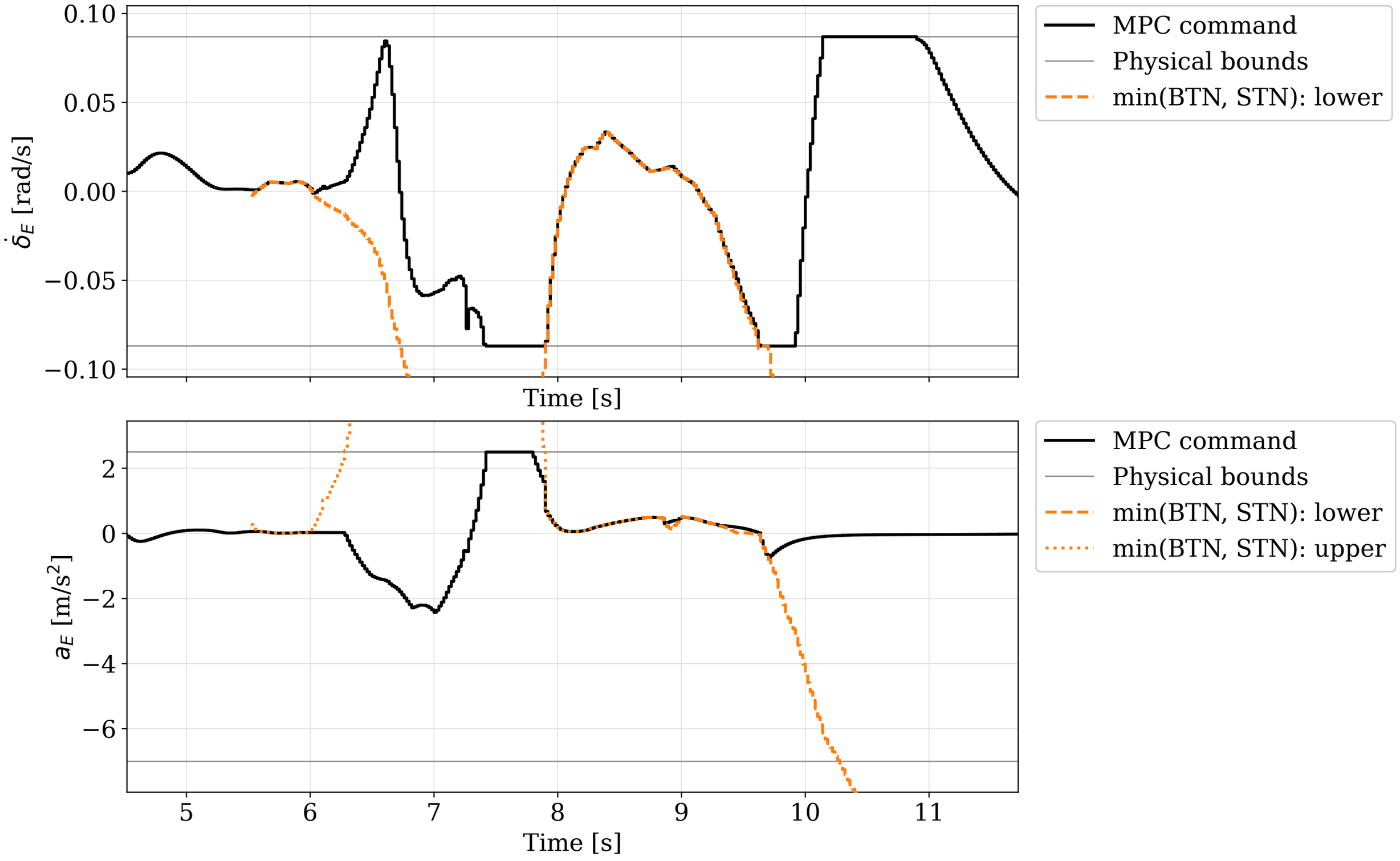

min(BTN, STN)
V
Time [s]
0
1
2
3
4
5
6
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5

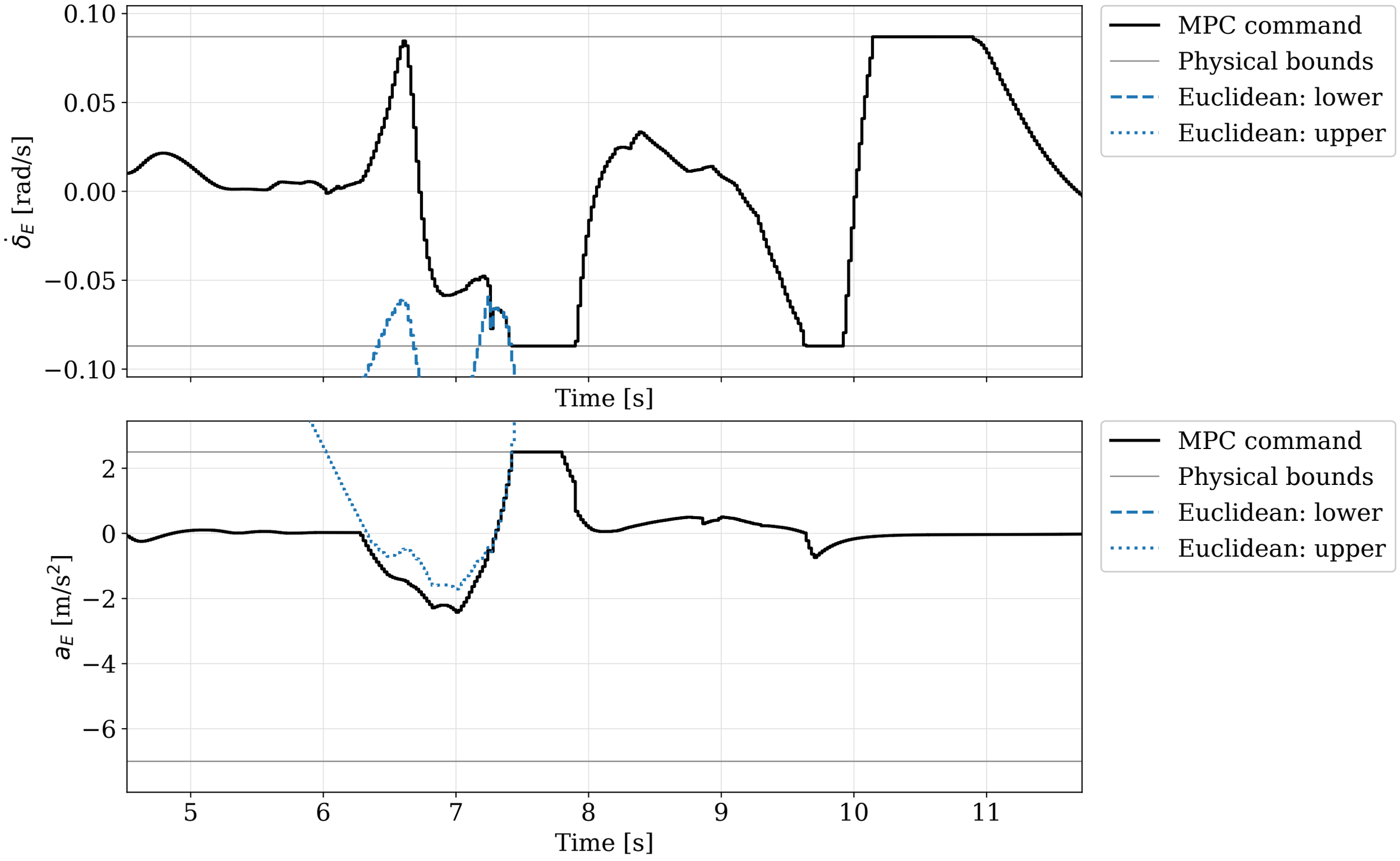

Euclidean
V
8
6
4
2
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time [s]

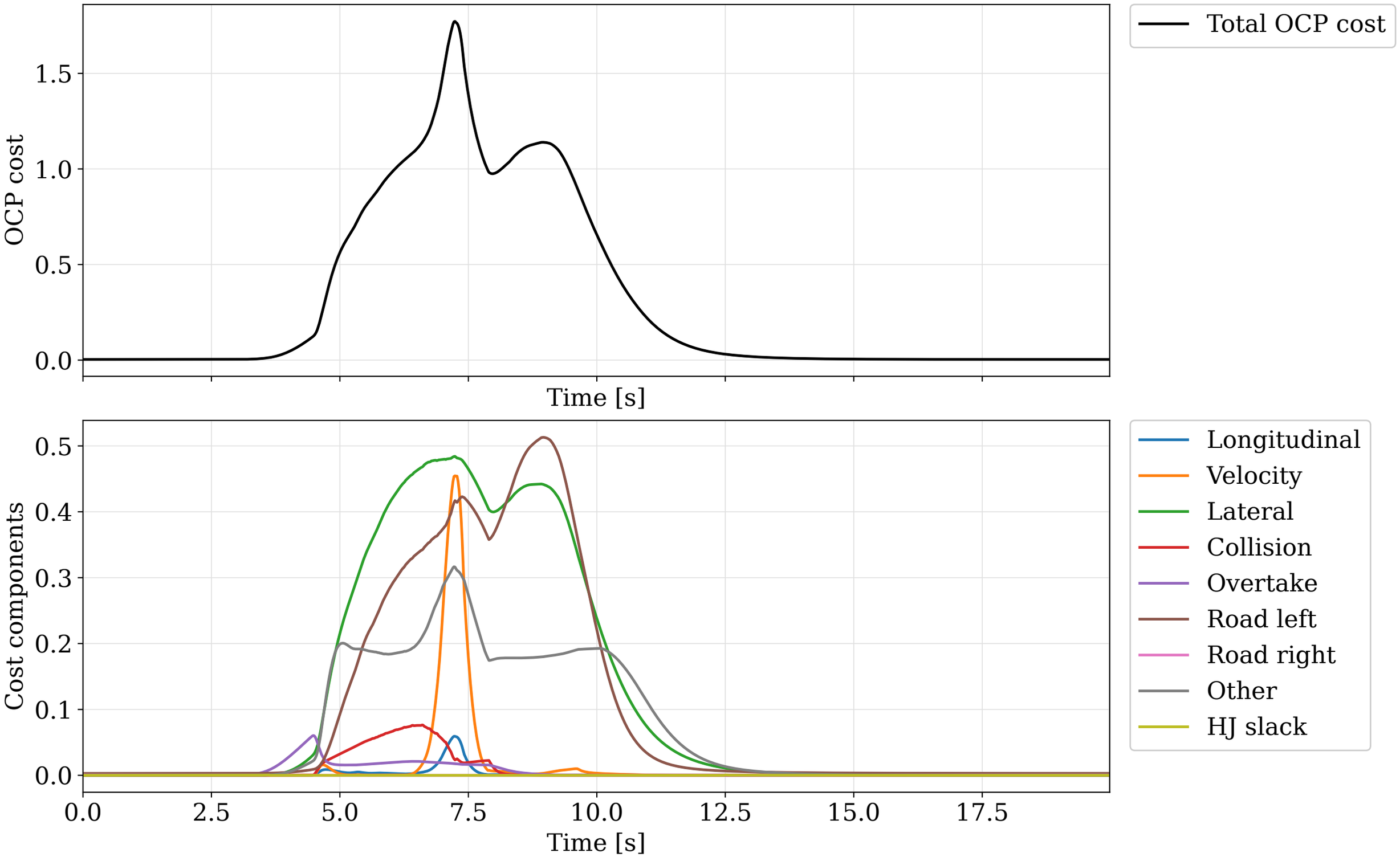

0.10
0.05
0.00
−0.05
−0.10
$\dot{\delta}_E$ [rad/s]
Time [s]
MPC command
Physical bounds
BTN: lower
STN: lower
STN: upper
min(BTN, STN): lower
Euclidean: lower
Euclidean: upper
2
0
−2
−4
−6
a_E [m/s^2]
5
6
7
8
9
10
11
Time [s]
MPC command
Physical bounds
BTN: lower
BTN: upper
STN: lower
STN: upper
min(BTN, STN): lower
min(BTN, STN): upper
Euclidean: lower
Euclidean: upper

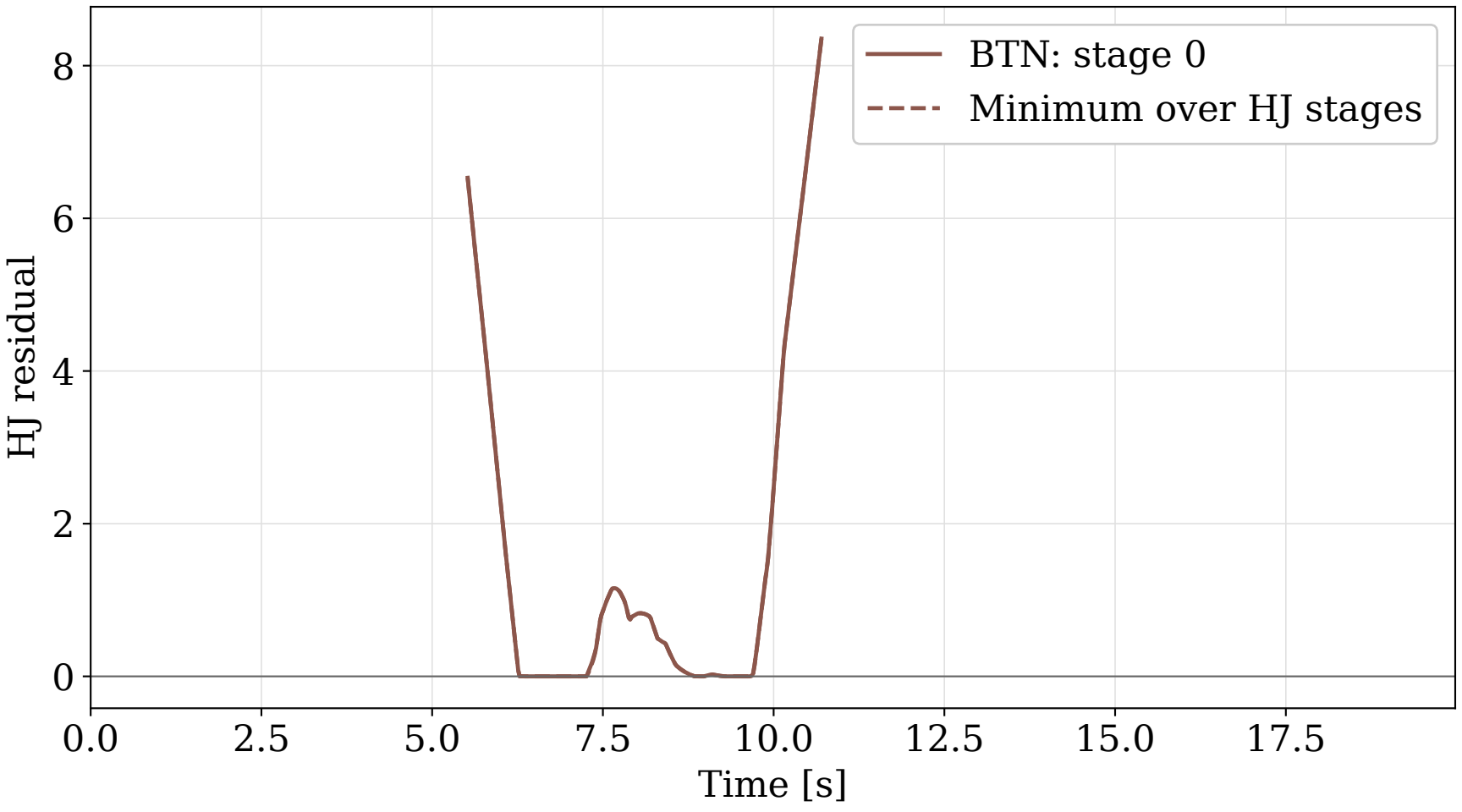

0.10
0.05
0.00
−0.05
−0.10
$\dot{\delta}_E$ [rad/s]
Time [s]
MPC command
Physical bounds
BTN: lower
2
0
−2
−4
−6
a_E [m/s^2]
5
6
7
8
9
10
11
Time [s]
MPC command
Physical bounds
BTN: lower
BTN: upper

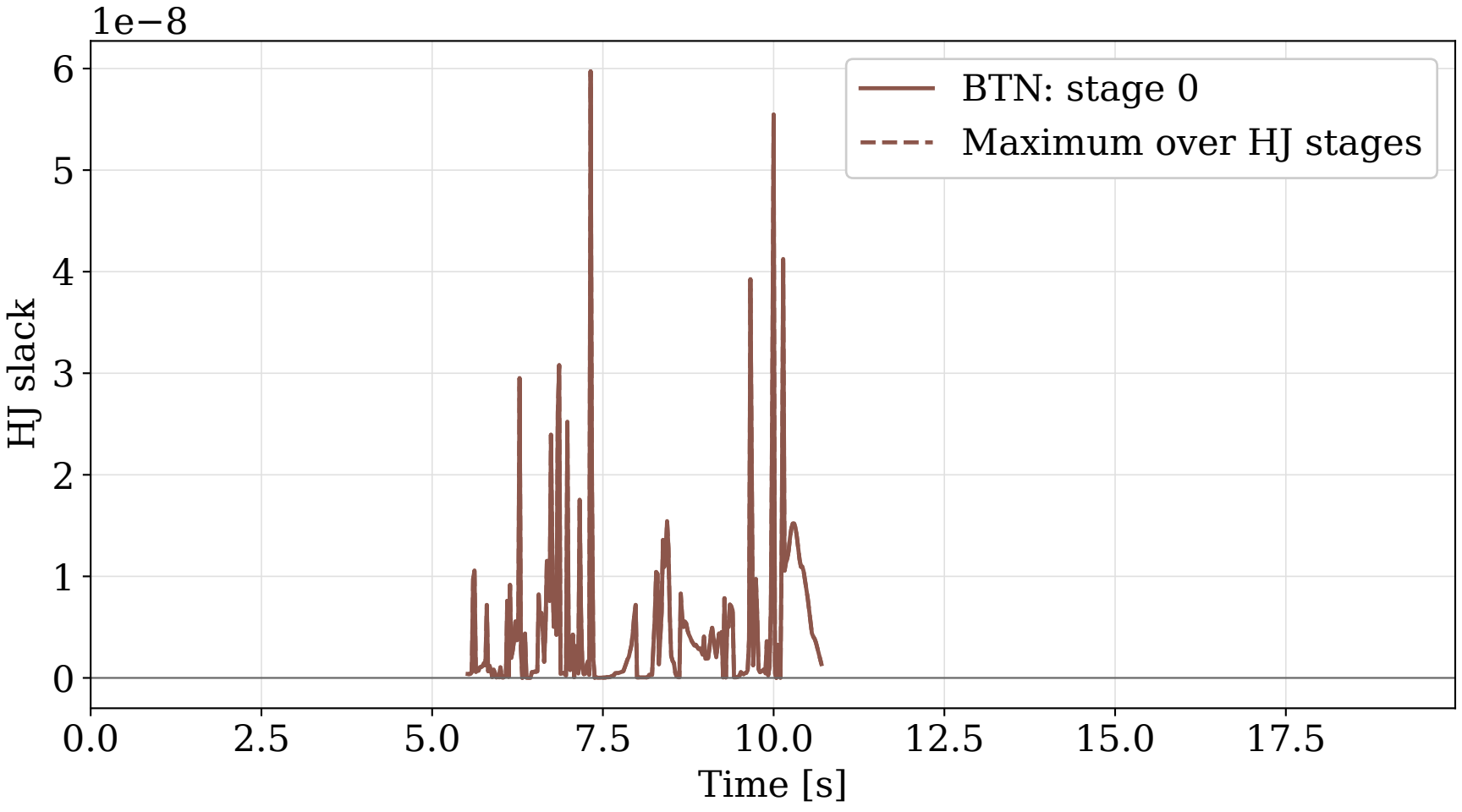

0.10
0.05
0.00
−0.05
−0.10
$\dot{\delta}_E$ [rad/s]
Time [s]
MPC command
Physical bounds
STN: lower
STN: upper
2
0
−2
−4
−6
a_E [m/s^2]
5
6
7
8
9
10
11
Time [s]
MPC command
Physical bounds
STN: lower
STN: upper

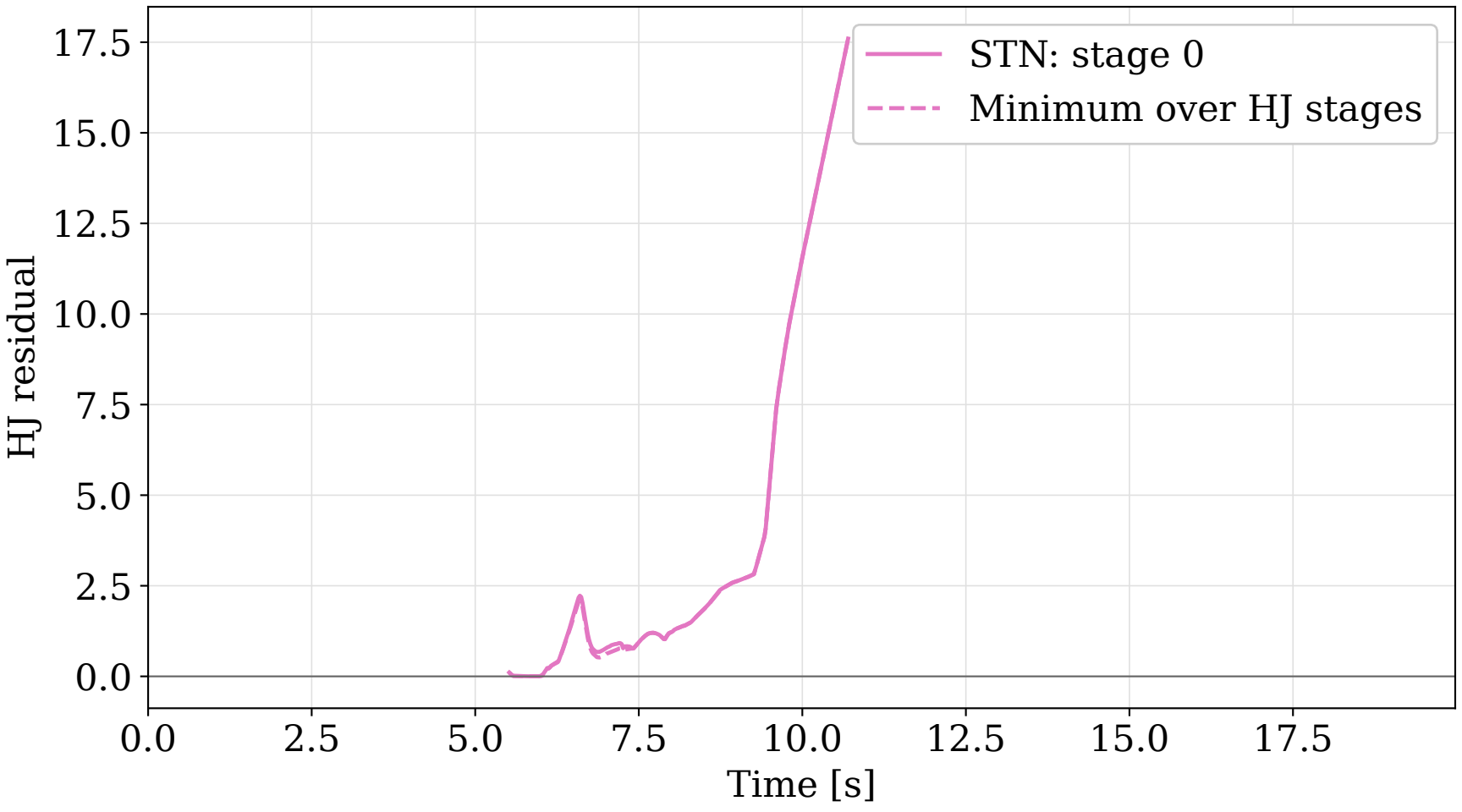

0.10
0.05
0.00
−0.05
−0.10
$\dot{\delta}_E$ [rad/s]
Time [s]
MPC command
Physical bounds
min(BTN, STN): lower
2
0
−2
−4
−6
a_E [m/s^2]
5
6
7
8
9
10
11
Time [s]
MPC command
Physical bounds
min(BTN, STN): lower
min(BTN, STN): upper

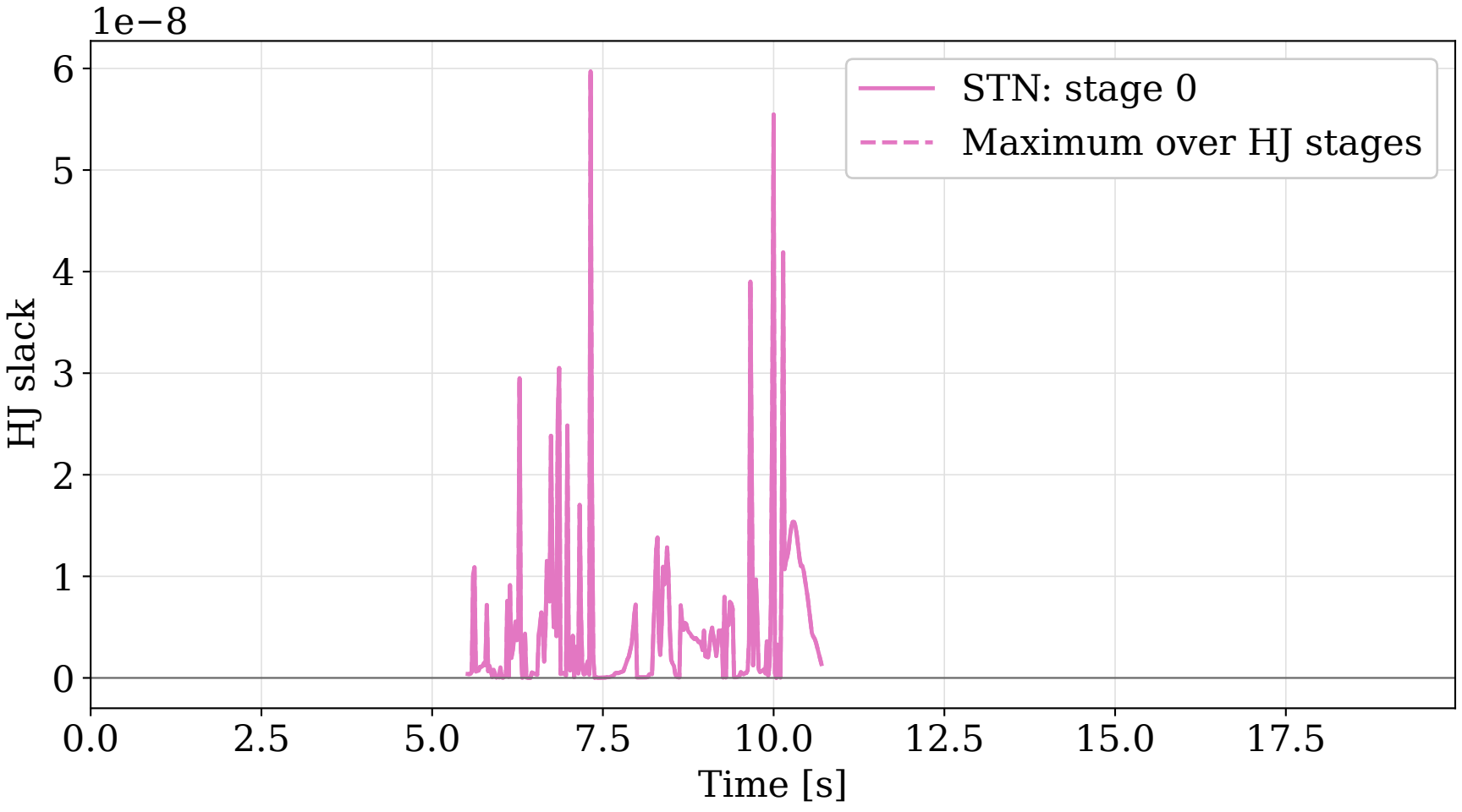

0.10
0.05
0.00
−0.05
−0.10
$\dot{\delta}_E$ [rad/s]
Time [s]
MPC command
Physical bounds
Euclidean: lower
Euclidean: upper
2
0
−2
−4
−6
a_E [m/s^2]
5
6
7
8
9
10
11
Time [s]
MPC command
Physical bounds
Euclidean: lower
Euclidean: upper

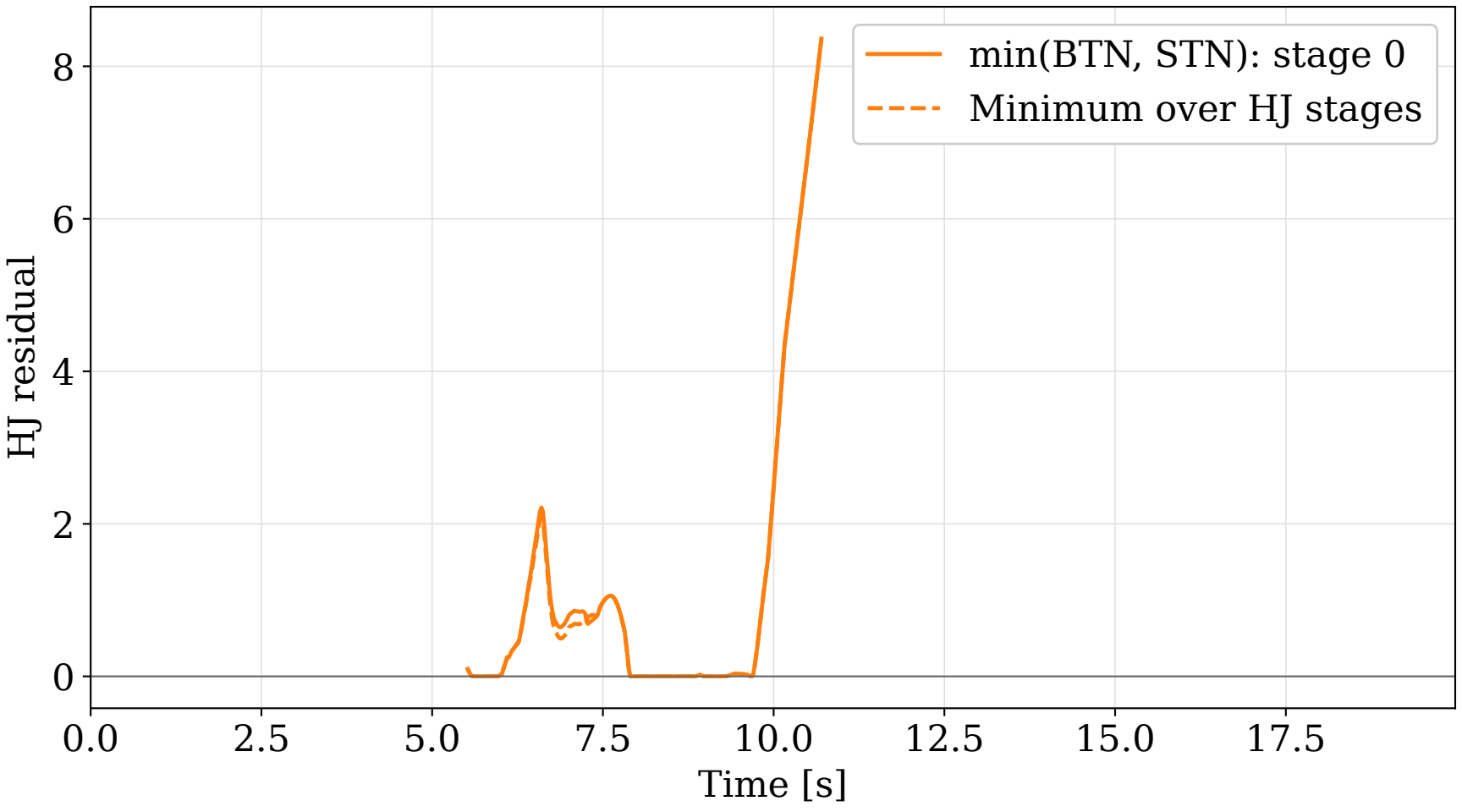

Total OCP cost
OCP cost
0.0
0.5
1.0
1.5
Time [s]
Longitudinal
Velocity
Lateral
Collision
Overtake
Road left
Road right
Other
HJ slack
Cost components
0.0
0.1
0.2
0.3
0.4
0.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time [s]

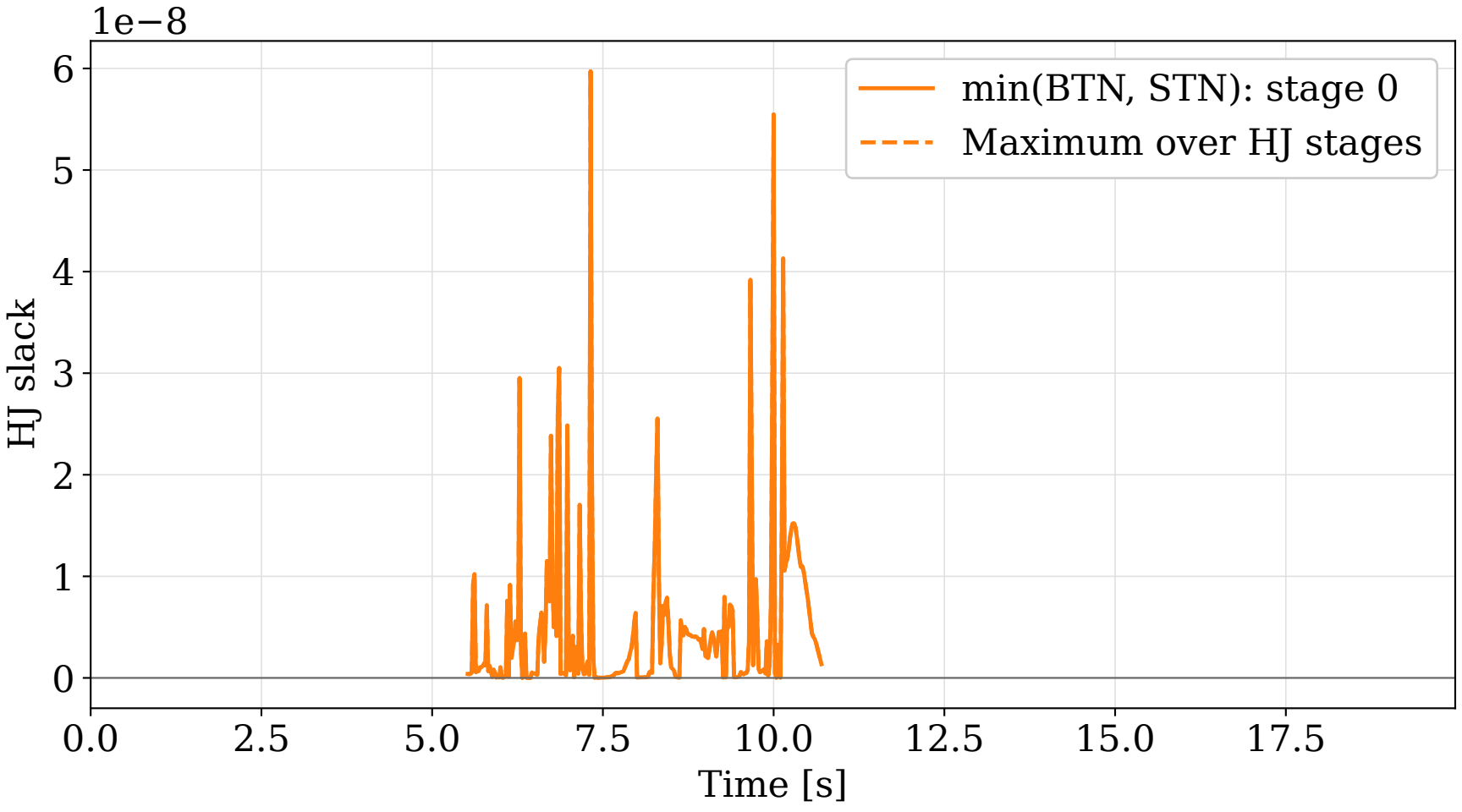

BTN: stage 0
Minimum over HJ stages
HJ residual
Time [s]
0
2
4
6
8
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5

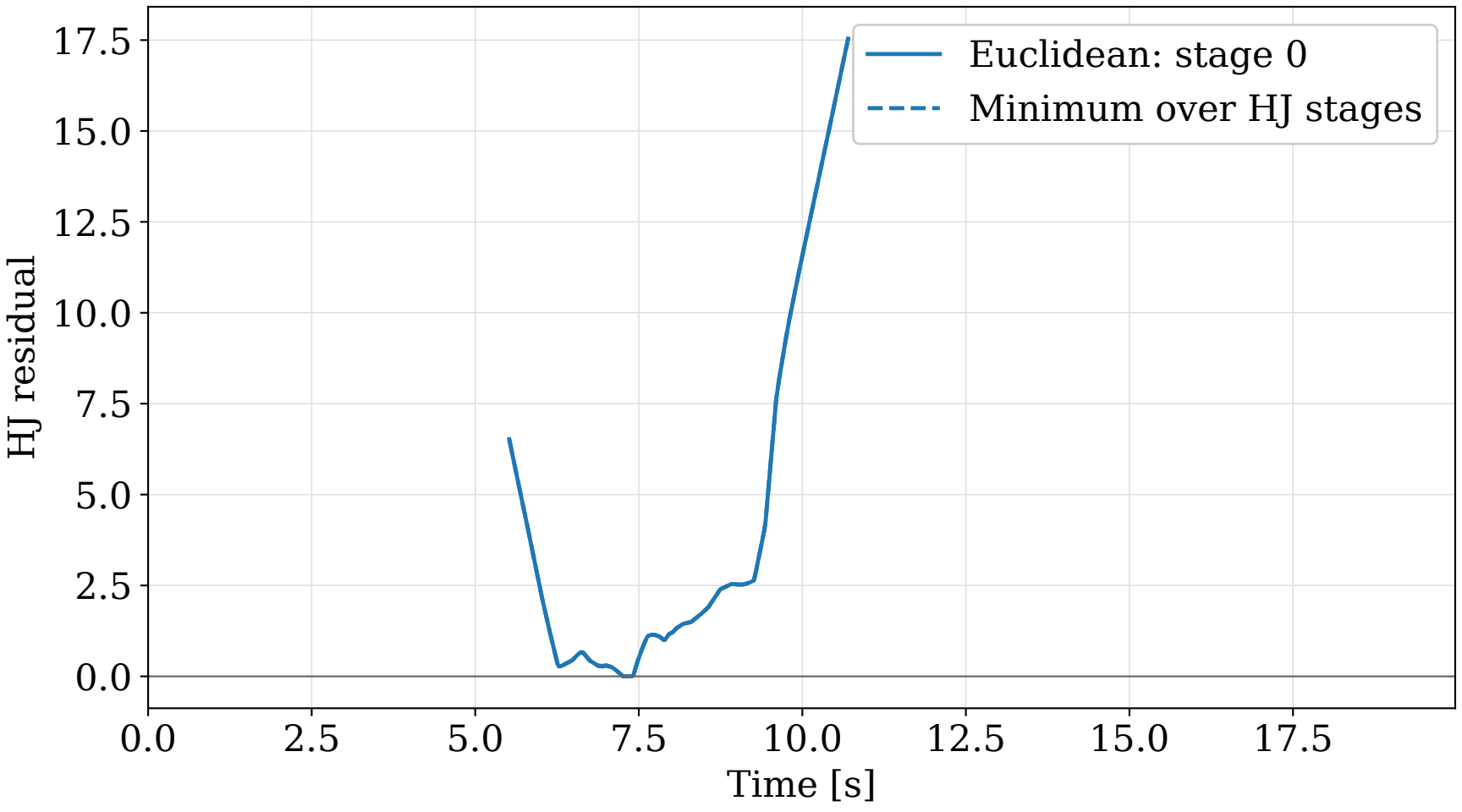

1e−8
6
5
4
3
2
1
0
HJ slack
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time [s]
BTN: stage 0
Maximum over HJ stages

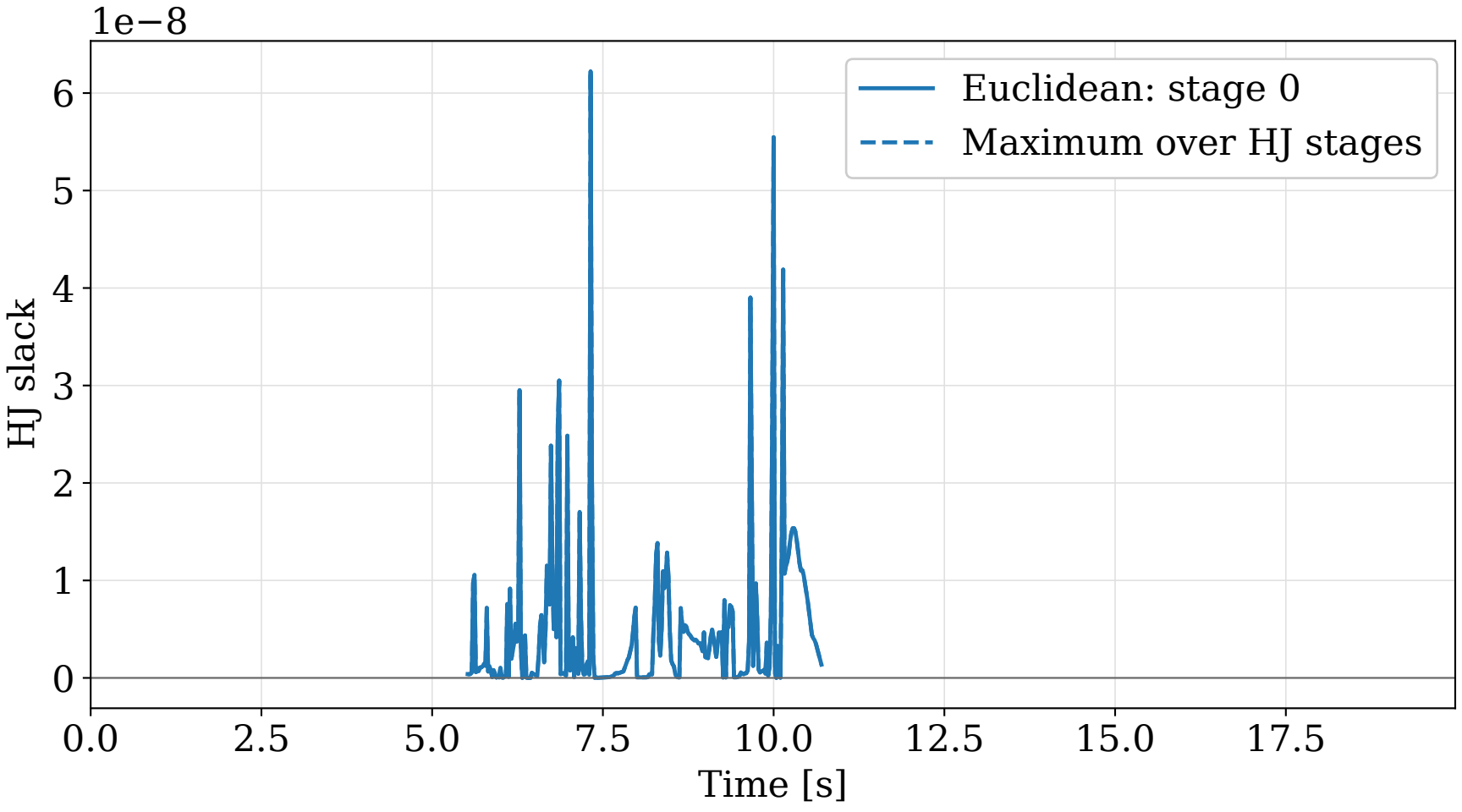

STN: stage 0
Minimum over HJ stages
HJ residual
Time [s]
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5

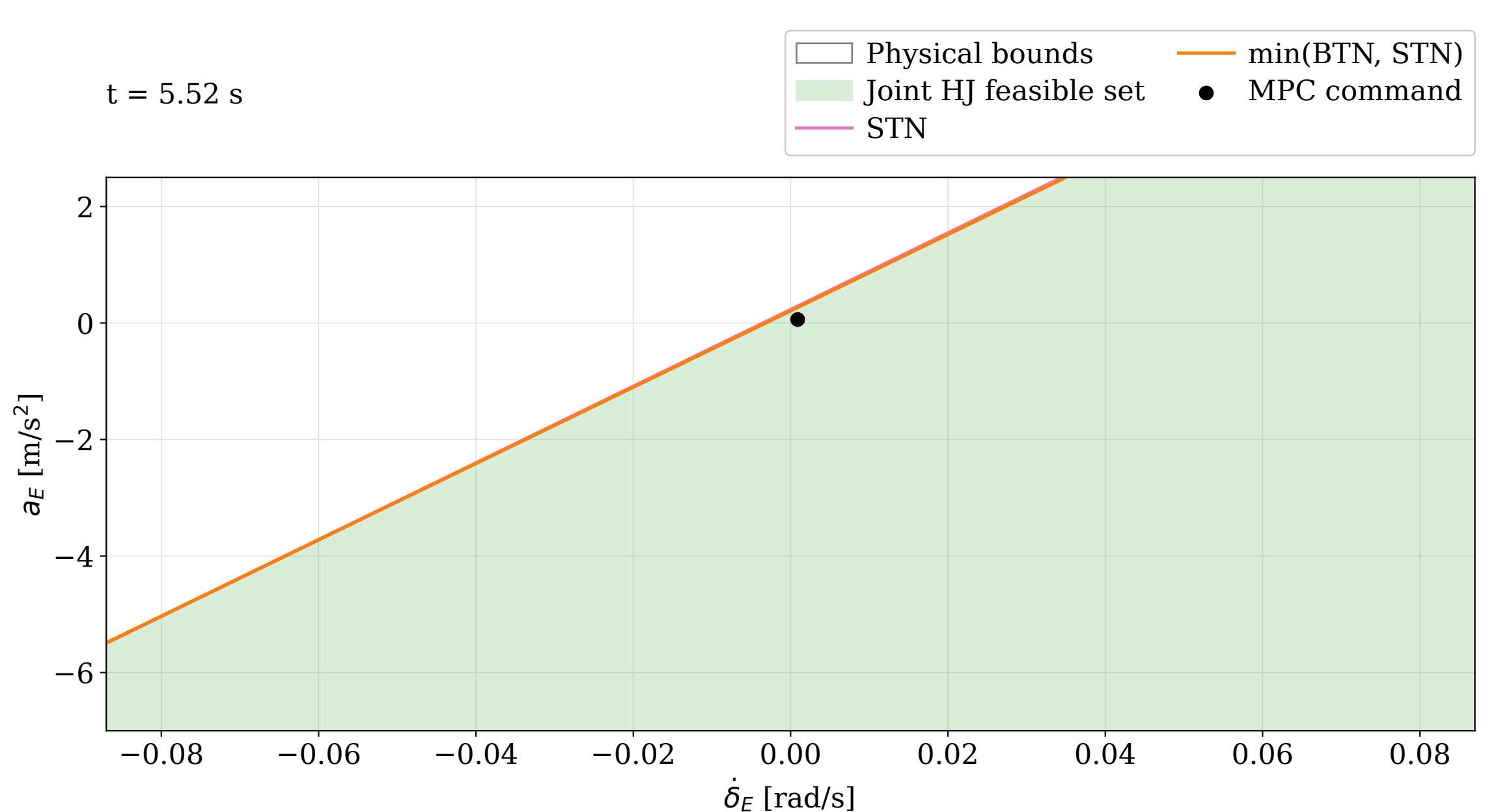

1e−8
6
5
4
3
2
1
0
HJ slack
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time [s]
STN: stage 0
Maximum over HJ stages

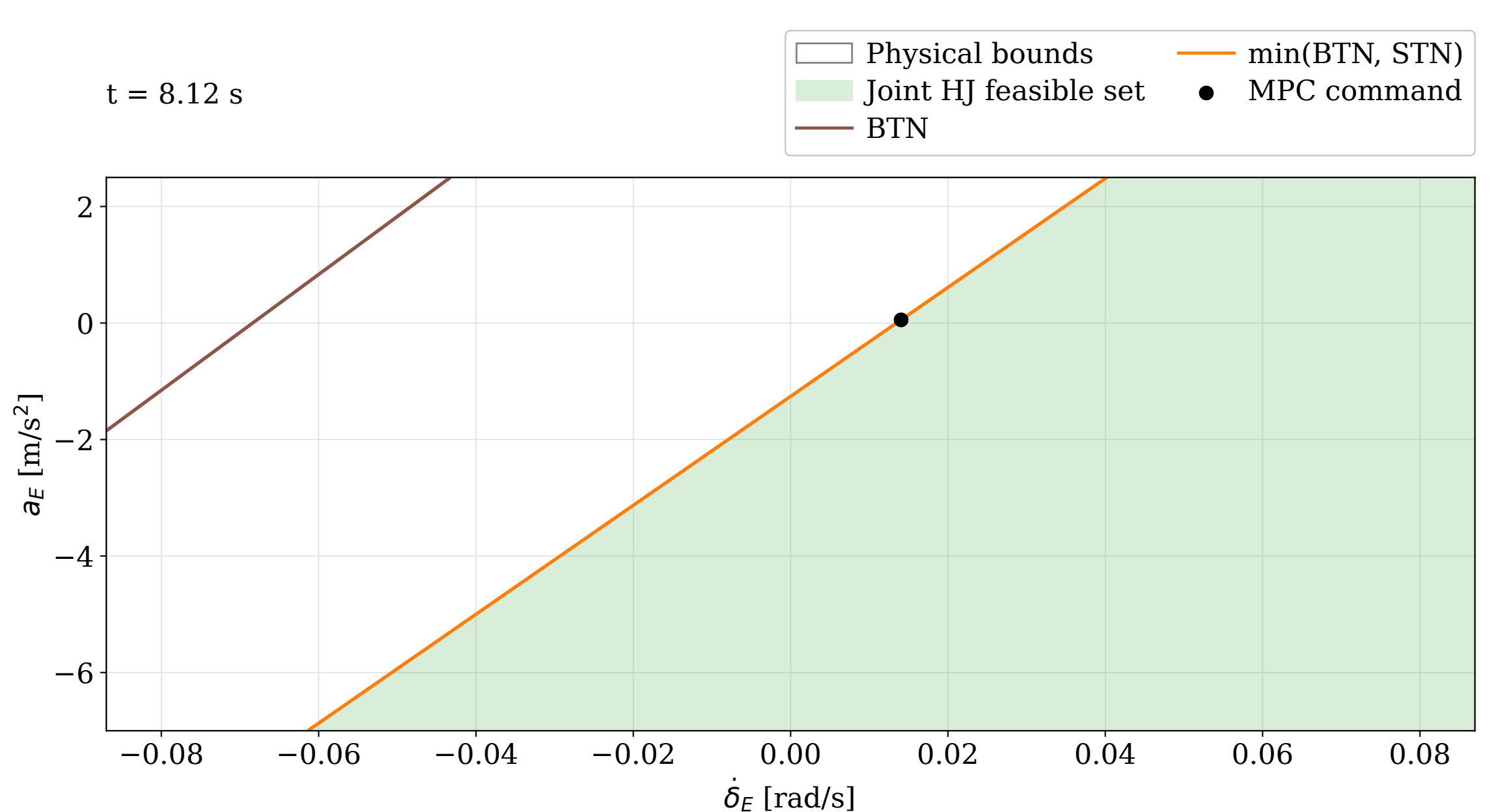

min(BTN, STN): stage 0
Minimum over HJ stages
HJ residual
Time [s]
0
2
4
6
8
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5

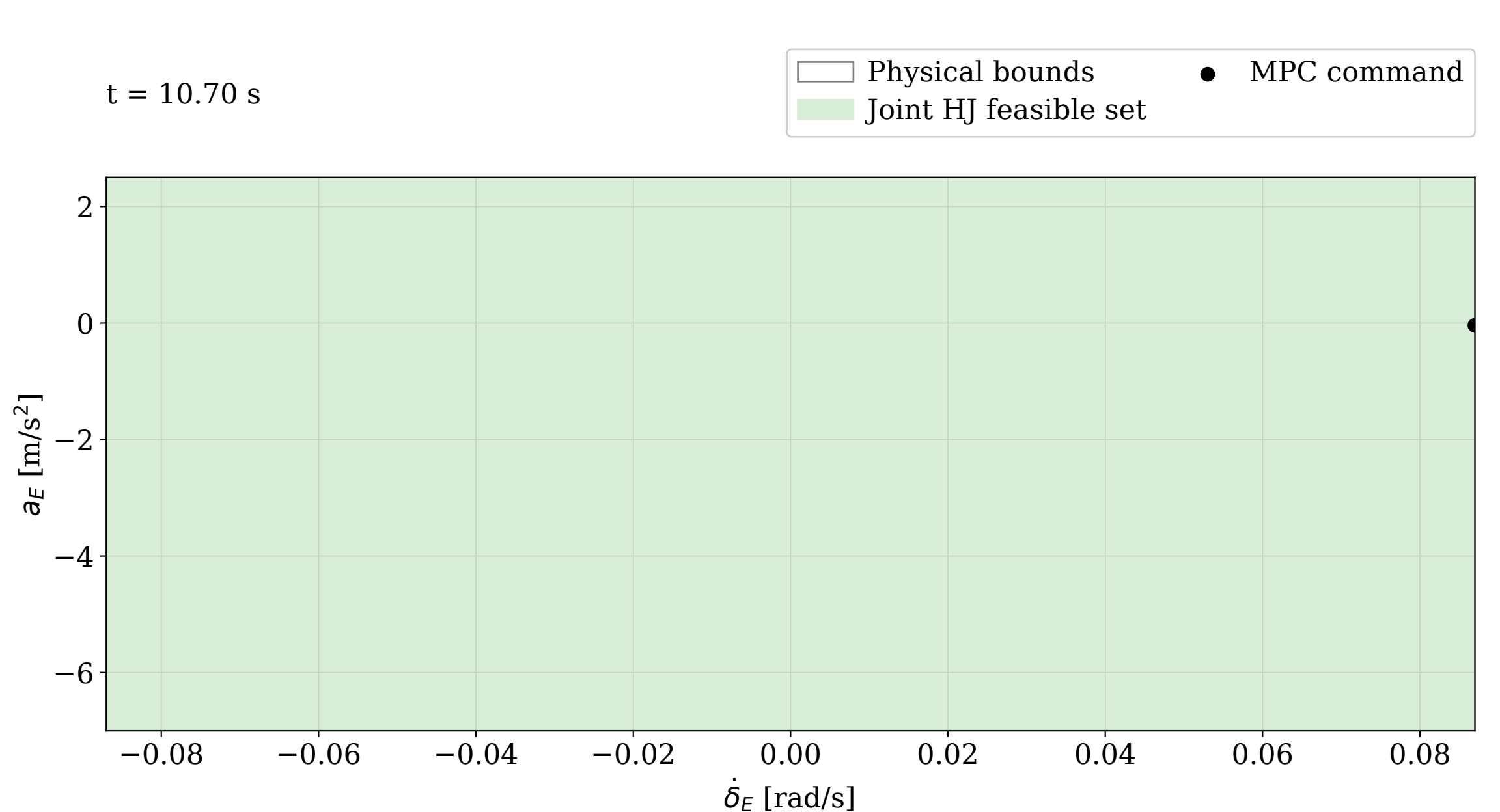

1e−8
6
5
4
3
2
1
0
HJ slack
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time [s]
min(BTN, STN): stage 0
Maximum over HJ stages


Euclidean: stage 0
Minimum over HJ stages
HJ residual
Time [s]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


1e−8
6
5
4
3
2
1
0
HJ slack
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time [s]
Euclidean: stage 0
Maximum over HJ stages


t = 5.52 s
Physical bounds
Joint HJ feasible set
STN
min(BTN, STN)
MPC command
2
0
−2
−4
−6
a_E [m/s^2]
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
$\dot{\delta}_E$ [rad/s]


t = 8.12 s
Physical bounds
Joint HJ feasible set
BTN
min(BTN, STN)
MPC command
2
0
−2
−4
−6
a_E [m/s^2]
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
$\dot{\delta}_E$ [rad/s]


t = 10.70 s
Physical bounds
Joint HJ feasible set
MPC command
2
0
−2
−4
−6
a_E [m/s^2]
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
$\dot{\delta}_E$ [rad/s]